

Jiarong (Bill) Lu

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RELEVANT SKILL

Language: C, C++, VHDL, Lean, Python, HTML, G-code

Tools: EDA/CAD — KiCad, Altium Designer, Autodesk Inventor, SolidWorks; Embedded — STM32 (STM32CubeIDE), FOC, PWM/ADC, JLCPCB; Software & AI — VS Code, GitHub, Docker, Cursor, Codex, Claude Code, Ollama, Dify, OpenClaw

EXPERIENCE

Studica Limited <i>Embedded Hardware & Firmware Developer</i>	Ontario, Canada <i>Jan 2026 - April 2026 Sep 2026 - Dec 2026 (Expected)</i>
- Designed and laid out a compact 24 V, 10 A+ brushless (BLDC) motor driver board only 3.5 cm in diameter, running field-oriented control (FOC). - Helped revise and modify a drone flight-controller board. - Leveraged AI tools to boost productivity; built a robot configurator that imports company products directly into SolidWorks/Inventor or runs as a standalone web app, letting users assemble custom robots and order in one click. - Developed and maintained firmware for five STM32-based products spanning motion, motor, distance, and color sensing. - Built an AI-assisted visualization tool for CNC-based sensor calibration, replacing manual G-code workflows.	
Waterloo Aerial Robotics Group (WARG) <i>Embedded Flight Software (EFS) Team Member</i>	Ontario, Canada <i>Oct 2024 - Oct 2026</i>
- Built the motor driver for the new driver architecture and the Attitude Manager with PID control - Modeled the thrust generated by the aircraft's motor(s) and propeller(s)	
Automate Agency <i>Web & AI Automation Developer</i>	Ontario, Canada <i>Part-time May 2026 - Present</i>
- Designed and launched automateagency.ca, a responsive website supporting service presentation and lead generation. - Built a daily Instagram workflow for post ideas, captions, creative assets, approval, scheduling, and publishing. - Developed an AI avatar video pipeline for recurring scripts and presenter videos. - Connected generation, review, scheduling, publishing, and logs into an end-to-end daily content system.	
Shanghai Mindmatrix Technology Ltd <i>Embedded Software Developer</i>	Shanghai, China <i>Jun 2025 - Sept 2025</i>
- Developed embedded firmware for BLDC motor control using STM32 microcontrollers, with real-time PWM, ADC, and communication via STM32CubeIDE and C/C++. - Designed and fabricated custom multi-layer BLDC driver PCBs using Altium Designer; managed manufacturing with JLCPCB. - Built a mechanical torque testing rig using Autodesk Inventor to evaluate motor performance under controlled load conditions. - Implemented advanced motor control algorithms, including Field-Oriented Control (FOC) and finely tuned PID loops, to achieve precise angular positioning with <0.1° error. - Successfully redesigned the PCB layout and reselected key components to achieve a compact 1.8 cm × 3.0 cm board, meeting strict integration requirements in constrained environments. Implemented STM32F446 microcontroller to enable precise brushless motor angle detection and three-phase output control.	

PROJECTS

Glory Media AI transformation	Remote Jun 2025 - Aug 2025
- Built a native iOS social app for media subscribers, enabling content sharing, interaction, and community engagement. - Developed a data platform aggregating influencer data from Instagram and LinkedIn, along with brand marketing data. - Applied AI models to analyze creator and campaign data, generating marketing recommendations for brands and media teams. - Explored enterprise deployment of OpenClaw by designing skills and knowledge bases, enabling AI agents to function as domain-specific experts. - Actively explored real-world AI applications by prototyping and evaluating AI-driven tools in content generation, automation, and decision support systems.	

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals Issued Jun 2025	Microsoft Certified: Azure Fundamentals Issued Jun 2025
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EDUCATION

Candidate for Bachelor of Applied Science, University of Waterloo Bachelor of Applied Science in Computer Engineering	Sept 2024 - Expected: June 2029
Shanghai Weiyu High School IBDP (International Baccalaureate Diploma Programme)	Sept 2021 - June 2024